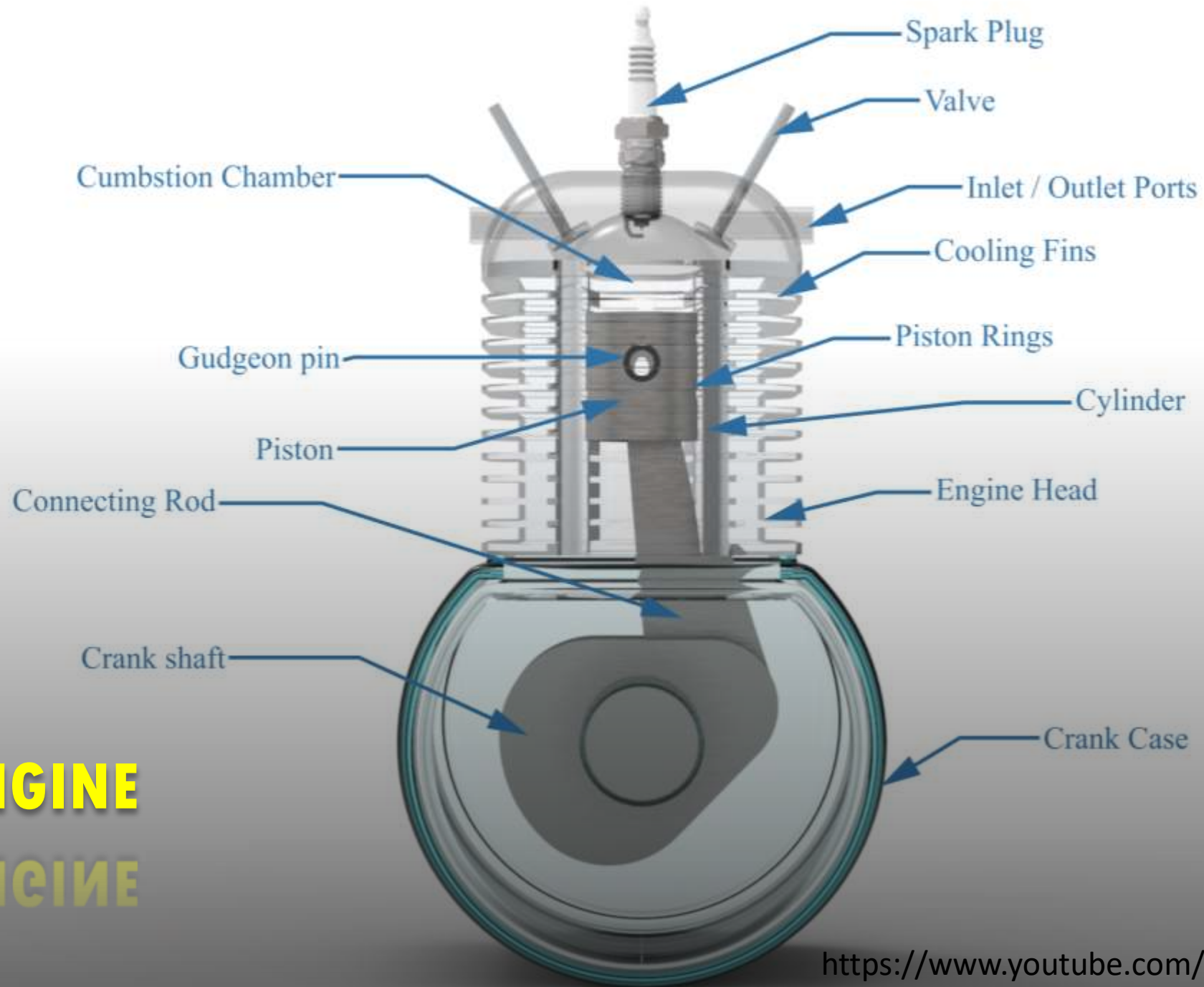


IC ENGINES

Application

1. Automotive:
 - (i) Car
 - (ii) Truck/Bus
 - (iii) Off-highway
2. Locomotive
3. Light Aircraft
4. Marine:
 - (i) Outboard
 - (ii) Inboard
 - (iii) Ship
5. Power Generation:
 - (i) Portable (Domestic)
 - (ii) Fixed (Peak Power)

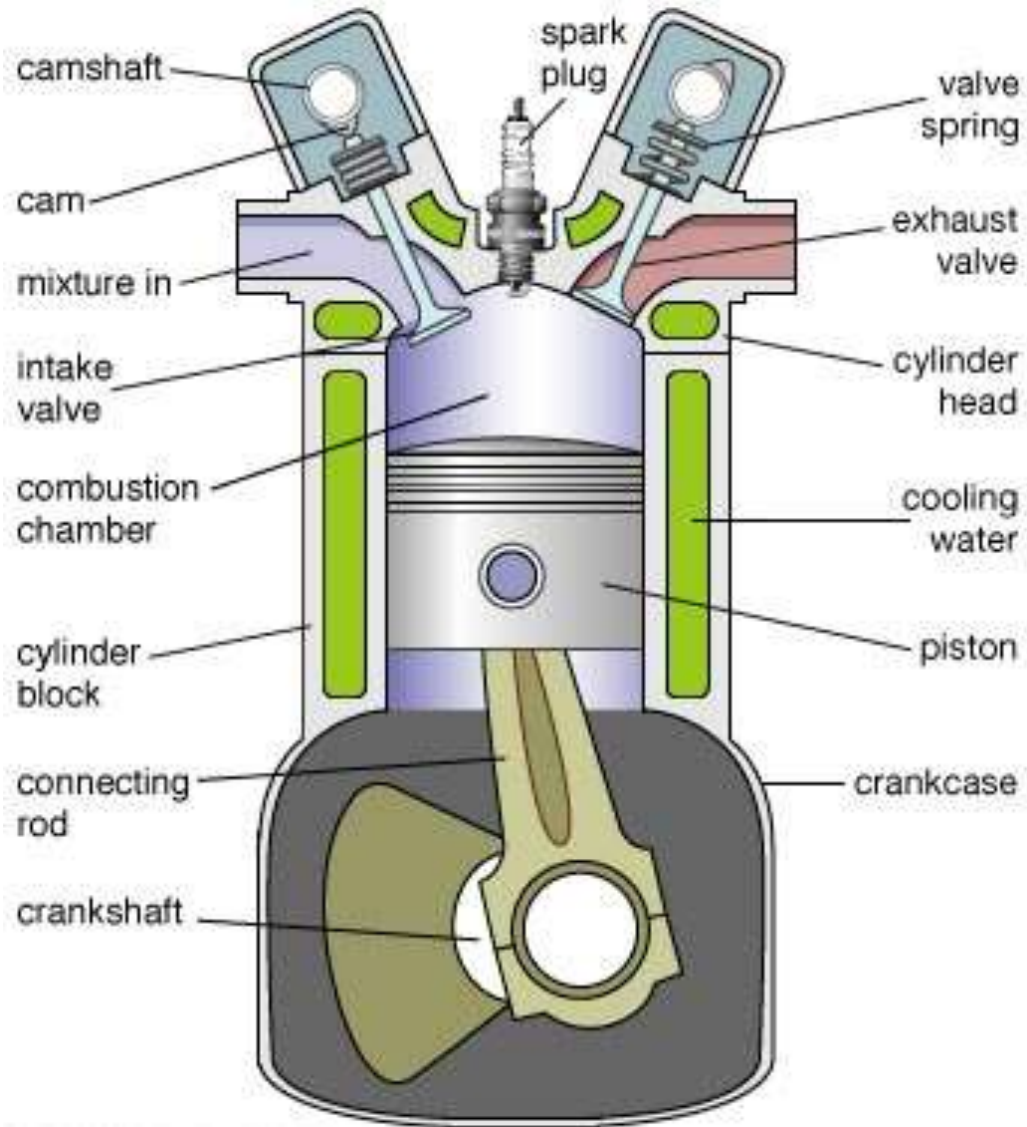




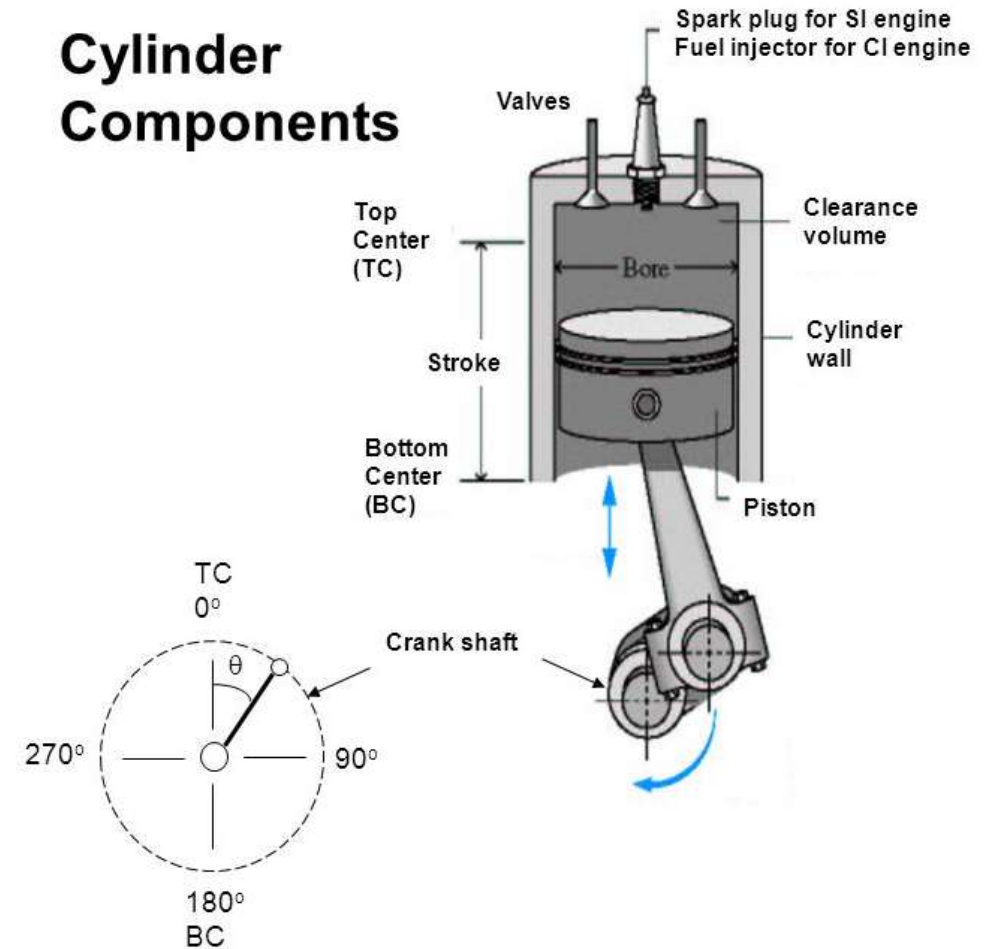
IC ENGINE

IC ENGINE

Main components of IC Engine



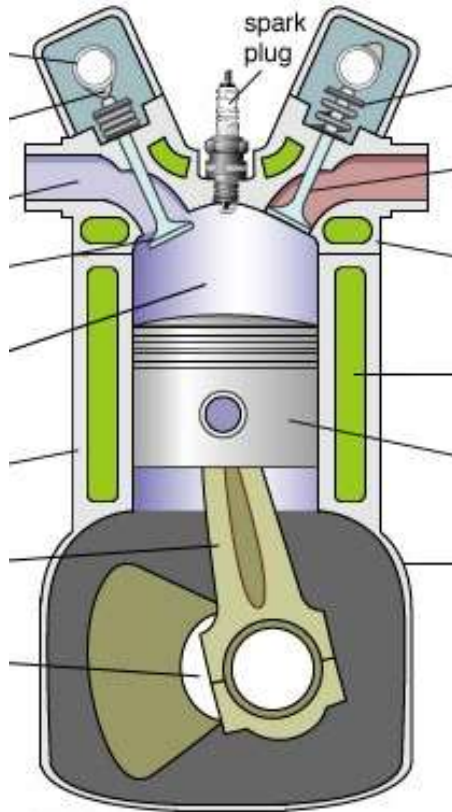
Cylinder Components



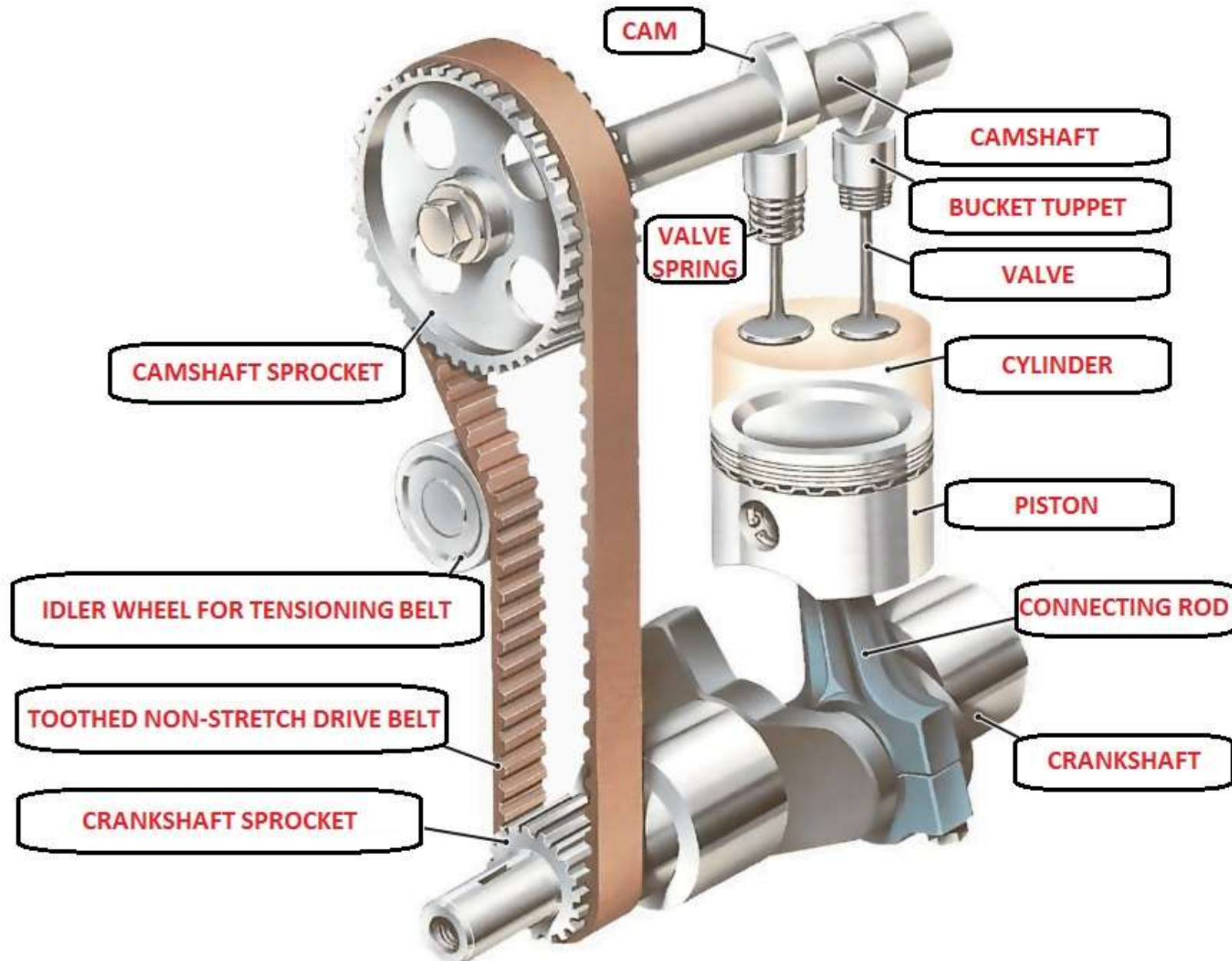
Main components of IC Engine



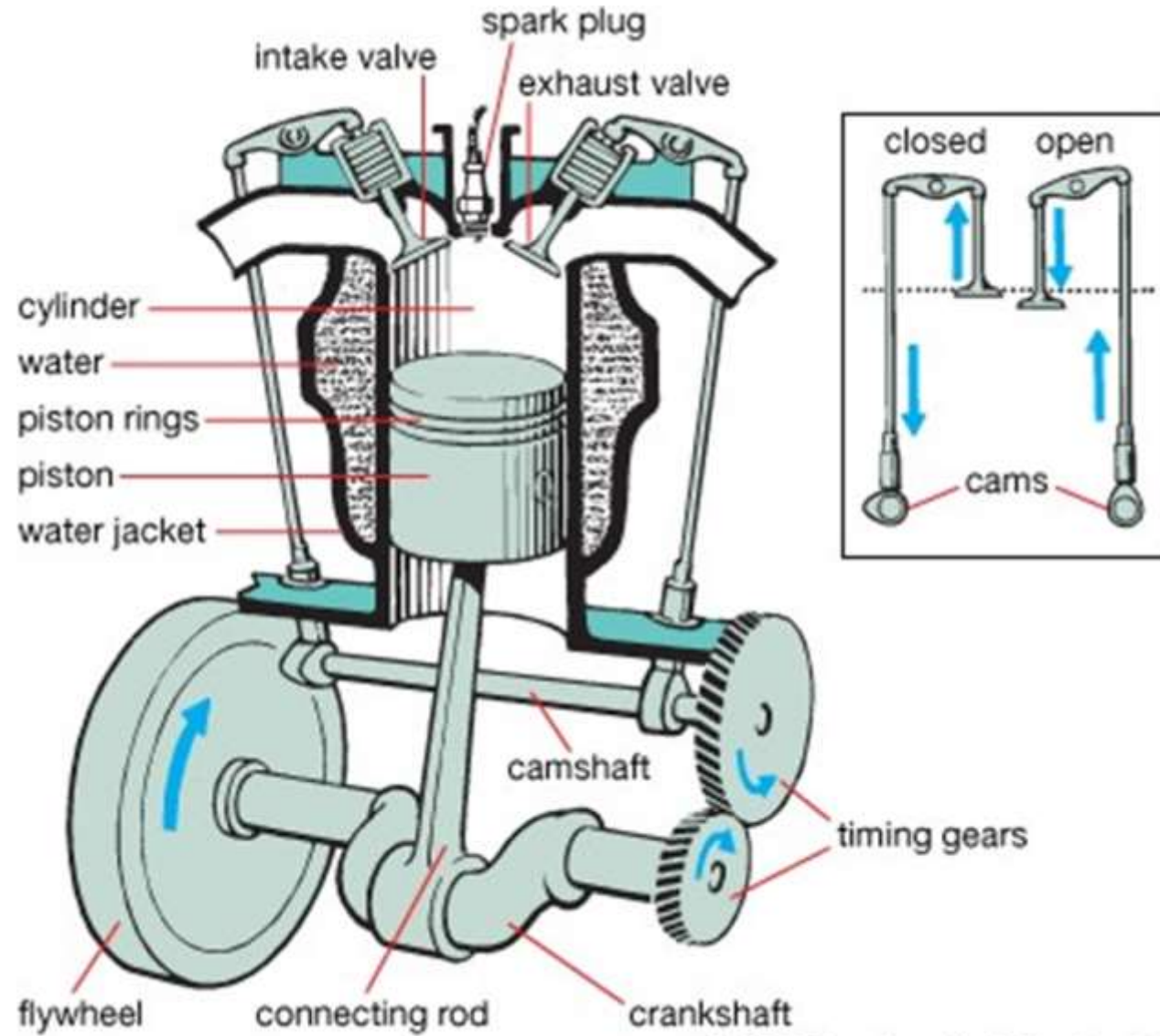
Main components of IC Engine



VALVE MECHANISM CONSTRUCTION



Engine Components



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Internal combustion Engine Components:

- I.C. Engine components shown in figure1 and figure2 are defined as follows:
- **Block** : Body of the engine containing cylinders, made of cast iron or aluminium.
- **Cylinder** : The circular cylinders in the engine block in which the pistons reciprocate back and forth.
- **Head** : The piece which closes the end of the cylinders, usually containing part of the clearance volume of the combustion chamber.
- **Combustion chamber**: The end of the cylinder between the head and the piston face where combustion occurs.
 - The size of combustion chamber continuously changes from minimum volume when the piston is at TDC to a maximum volume when the piston at BDC.

Internal combustion Engine Components:

- **Crankshaft** : Rotating shaft through which engine work output is supplied to external systems.
 - The crankshaft is connected to the engine block with the main bearings.
 - It is rotated by the reciprocating pistons through the connecting rods connected to the crankshaft, offset from the axis of rotation. This offset is sometimes called crank throw or crank radius.
- **Connecting rod** : Rod connecting the piston with the rotating crankshaft, usually made of steel or alloy forging in most engines but may be aluminum in some small engines.
- **Piston rings**: Metal rings that fit into circumferential grooves around the piston and form a sliding surface against the cylinder walls.

Internal combustion Engine Components:

- **Camshaft** : Rotating shaft used to push open valves at the proper time in the engine cycle, either directly or through mechanical or hydraulic linkage (push rods, rocker arms, tappets) .
- **Push rods** : The mechanical linkage between the camshaft and valves on overhead valve engines with the camshaft in the crankcase.
- **Crankcase** : Part of the engine block surrounding the crankshaft.
 - In many engines the oil pan makes up part of the crankcase housing.
- **Exhaust manifold** : Piping system which carries exhaust gases away from the engine cylinders, usually made of cast iron .

Internal combustion Engine Components:

- **Intake manifold** :Piping system which delivers incoming air to the cylinders, usually made of cast metal, plastic, or composite material.
 - In most SI engines, fuel is added to the air in the intake manifold system either by fuel injectors or with a carburetor.
 - The individual pipe to a single cylinder is called runner.
- **Carburetor** : A device which meters the proper amount of fuel into the air flow by means of pressure differential.
 - For many decades it was the basic fuel metering system on all automobile (and other) engines.
- **Spark plug** : Electrical device used to initiate combustion in an SI engine by creating high voltage discharge across an electrode gap.

Internal combustion Engine Components:

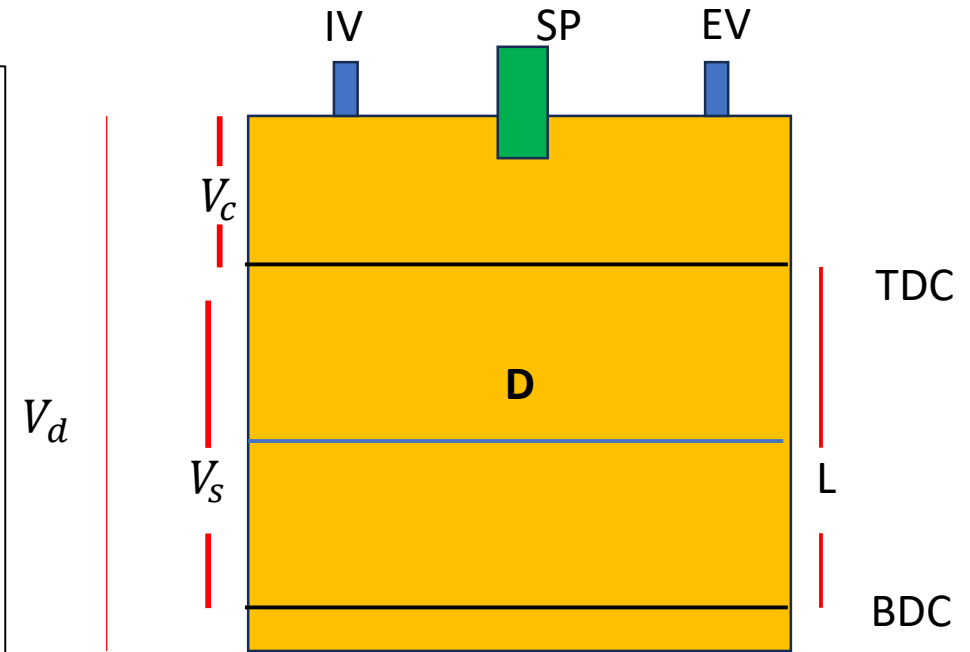
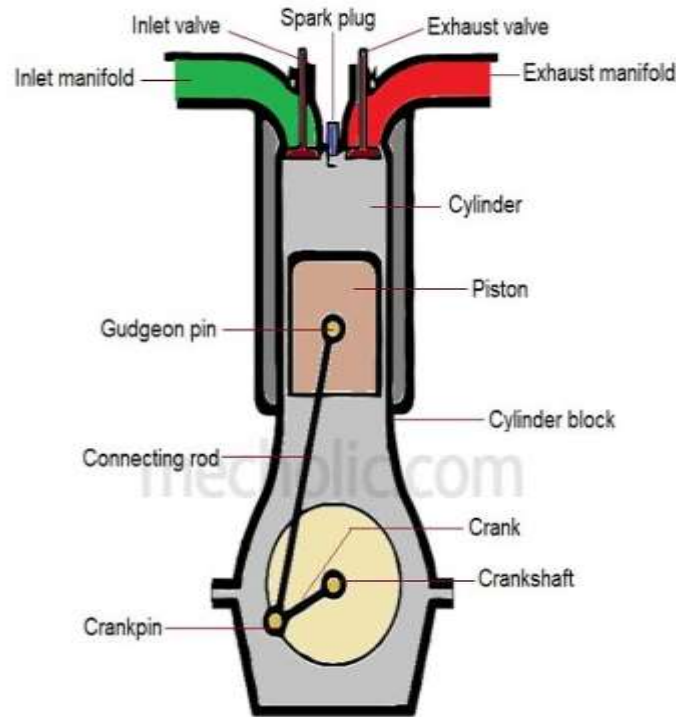
- **Exhaust System:** Flow system for removing exhaust gases from the cylinders, treating them, and exhausting them to the surroundings.
 - It consists of an exhaust manifold which carries the exhaust gases away from the engine,
 - a thermal or catalytic converter to reduce emissions, a muffler to reduce engine noise, and
 - a tailpipe to carry the exhaust gases away from the passenger compartment.
- **Flywheel :** Rotating mass with a large moment of inertia connected to the crank shaft of the engine.
 - The purpose of the flywheel is to store energy and furnish large angular momentum that keeps the engine rotating between power strokes and smooths out engine operation.

Internal combustion Engine Components:

- **Fuel injector** : A pressurized nozzle that sprays fuel into the incoming air (SI engines) or into the cylinder (CI engines).
- **Fuel pump** : Electrically or mechanically driven pump to supply fuel from the fuel tank (reservoir) to the engine.
- **Glow plug** : Small electrical resistance heater mounted inside the combustion chamber of many CI engines, used to preheat the chamber enough so that combustion will occur when first starting a cold engine.
 - The glow plug is turn off after the engine is started.
- **Starter** : Several methods are used to start IC engines. Most are started by use of an electric motor (starter) geared to the engine flywheel. Energy is supplied from an electric battery.

Terminology used in IC Engine

- 1.Cylinder Block
- 2.Cylinder
- 3.Piston
- 4.Combustion Chamber
- 5.Inlet Manifold
- 6.Exhaust Manifold
- 7.Inlet & Exhaust valve
- 8.Spark Plug
- 9.Connecting Rod
- 10.Crankshaft
- 11.Piston Ring
- 12.Gudgeon Pin
- 13.Camsaft



$$r = \frac{\text{cylinder volume at BDC}}{\text{cylinder volume at TDC}}$$

$$r = \frac{\text{cylinder volume} + \text{Cylinder clearance Volume}}{\text{cylinder clearnace Volume}}$$

$$r = \frac{V_s + V_c}{V_c}$$

$$\text{Displacement or stroke volume } (V_s) = A \times L = \frac{\pi}{4} D^2 L$$

$$\text{Cylinder volume}(V) = \text{displacement volume} + \text{clearance volume} = V_s + V_c$$

$$\text{Cylinder volume}(V) = V_s + V_c$$

1. Bore (d)

3. Bottom Dead Center BDC

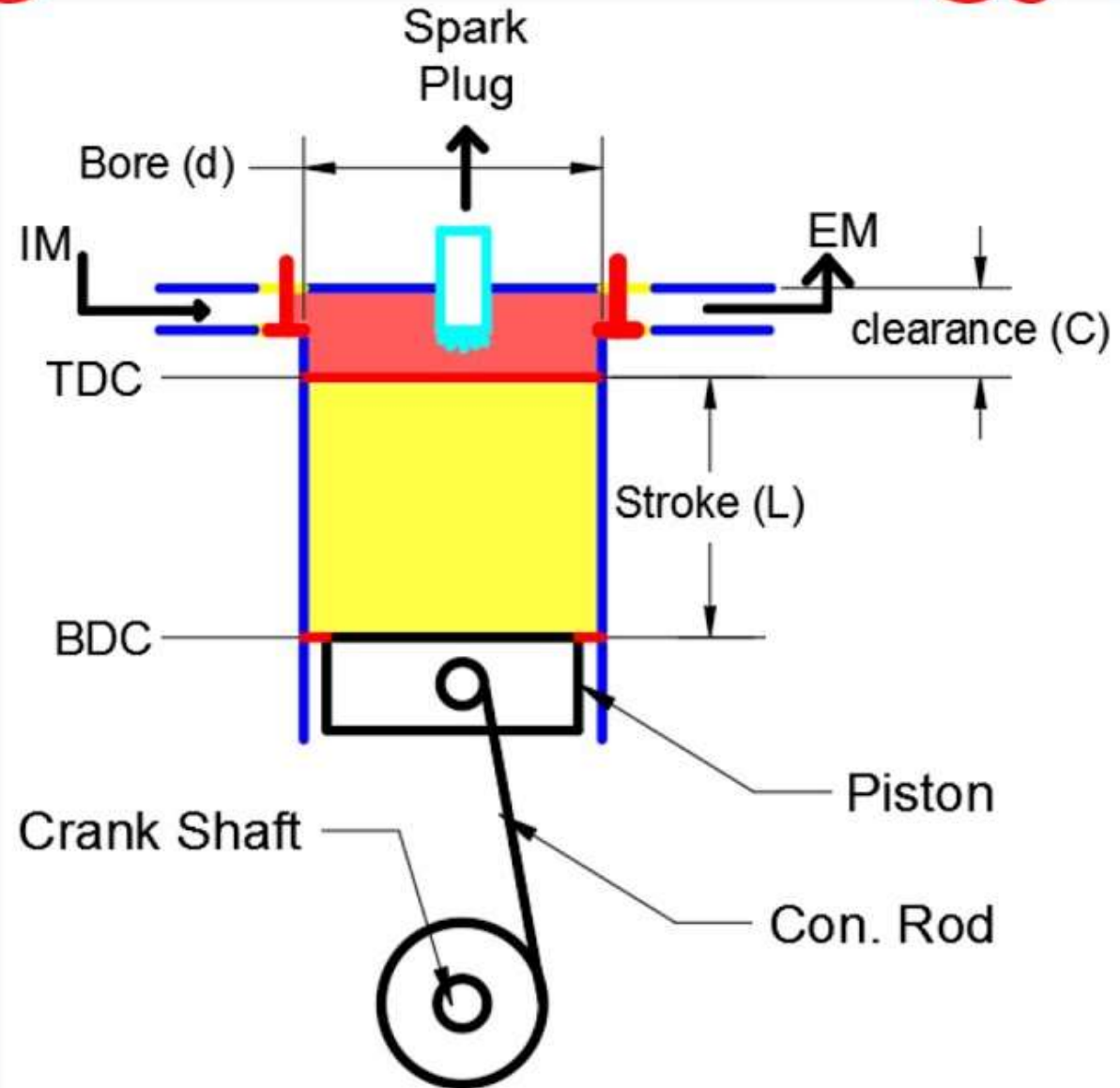
5. Swept Volume (V_s)

6. Clearance Volume (V_c)

7. Engine Capacity (CC)

8. Stroke to Bore ratio (L/D)

9. Compression Ratio (r)



Engine Terminology

- **Top Dead Center (TDC):** Position of the piston when it stops at the furthest point away from the crankshaft.
 - When the piston is at TDC, the volume in the cylinder is a minimum called the clearance volume.
- **Bottom Dead Center (BDC):** Position of the piston when it stops at the point closest to the crankshaft.
 - Some sources call this **Crank End Dead Center (CEDC)** because it is not always at the bottom of the engine. Some source call this point **Bottom Center (BC)**.
- **Stroke :** Distance traveled by the piston from one extreme position to the other : TDC to BDC or BDC to TDC.

Engine Terminology

- **Bore** :It is defined as cylinder diameter or piston face diameter; piston face diameter is same as cylinder diameter(minus small clearance).
- **Swept volume/Displacement volume** : Volume displaced by the piston as it travels through one stroke.
 - Swept volume is defined as stroke times bore.
 - Displacement can be given for one cylinder or entire engine (one cylinder times number of cylinders).

Engine Terminology

- **Clearance volume** : It is the minimum volume of the cylinder available for the charge (air or air fuel mixture) when the piston reaches at its outermost point (top dead center or outer dead center) during compression stroke of the cycle.
 - Minimum volume of combustion chamber with piston at TDC.
- **Compression ratio (r)** : The ratio of total volume to clearance volume of the cylinder is the compression ratio of the engine.
 - Typically, compression ratio for **SI** engines varies form **8 to 12** and for **CI** engines it varies from **12 to 24**

Types of IC engine

- **Spark Ignition engine** (SI Engine, **Petrol or Carburetor type**)
 - **Two stroke** – Two piston strokes over one revolution for one cycle to produce power.
 - **Four stroke** – Four piston strokes over two revolutions for one cycle to produce power.
- **Compression Ignition engine** (CI Engine, **Diesel type**)
 - Same as above for two and four stroke engine

Four strokes engine

4 - Stroke SI engine working principle

Movement of the piston up or down the cylinder makes up one stroke.

It is the distance between TDC and BDC and is equal to twice the radius of the crank.

In the four stroke engine cycle there are four strokes namely,

- ♦ Suction or Intake stroke,
- ♦ Compression stroke,
- ♦ Expansion stroke or power stroke,
- ♦ exhaust stroke.

One cycle completes when piston moves for four strokes or two revolutions of the crankshaft.

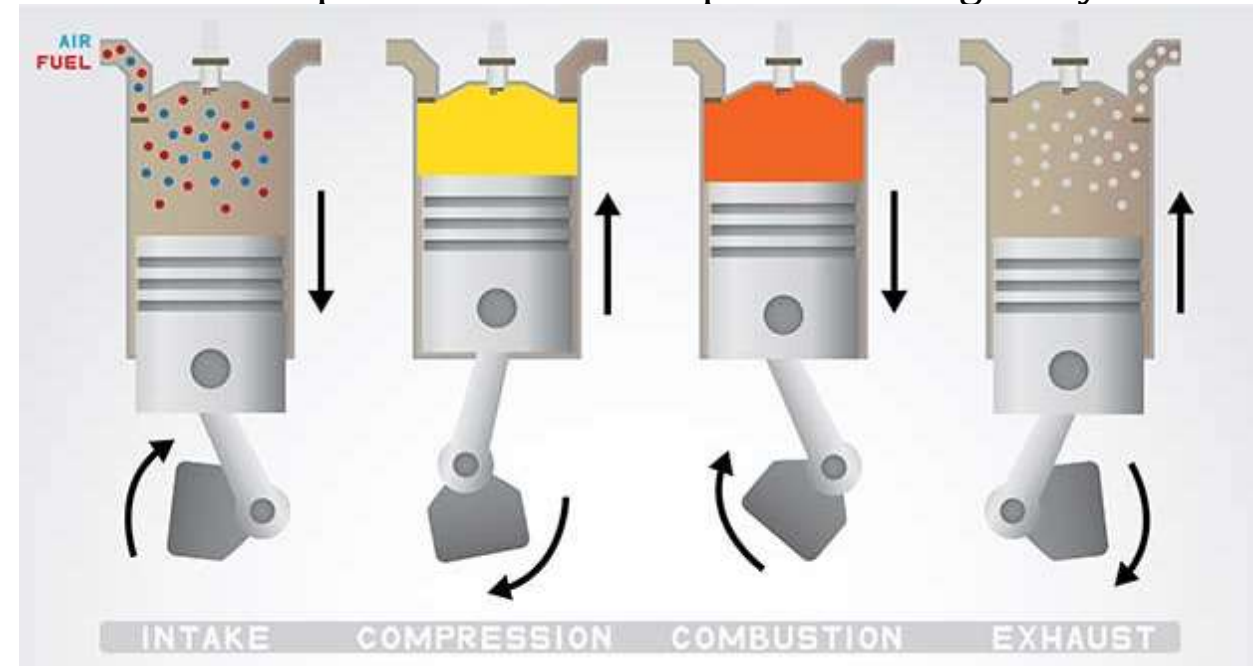
There are four strokes which are as follows:

- i) **Suction stroke** - Intake of air fuel mixture in cylinder through intake manifold.
- ii) **Compression stroke** - This compresses air fuel mixture, raising both the pressure and temperature in the cylinder. Near the end of the compression stroke, the spark plug is fired and the combustion is initiated. This combustion is almost an explosion, so this is assumed to be constant volume heat addition.
- iii) **Expansion or working or power stroke** - This is the stroke which produces work output of the engine cycle.

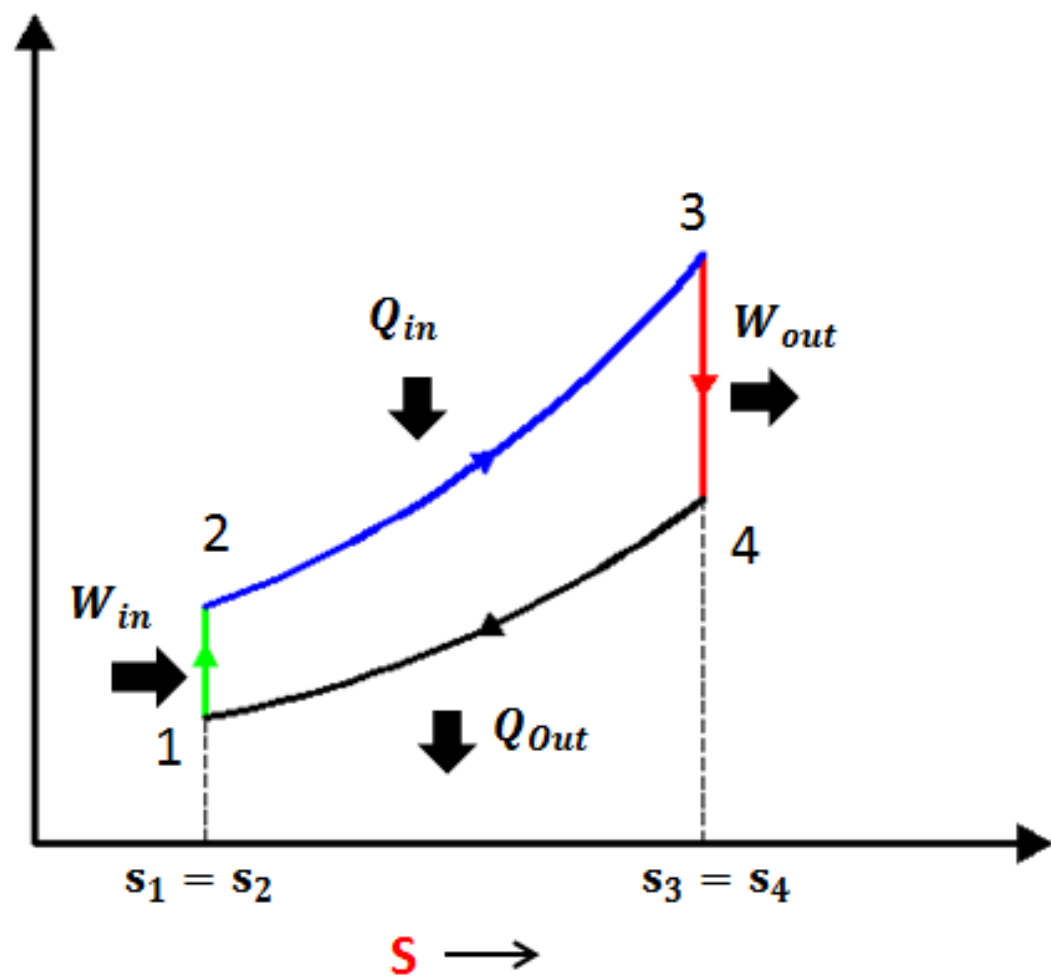
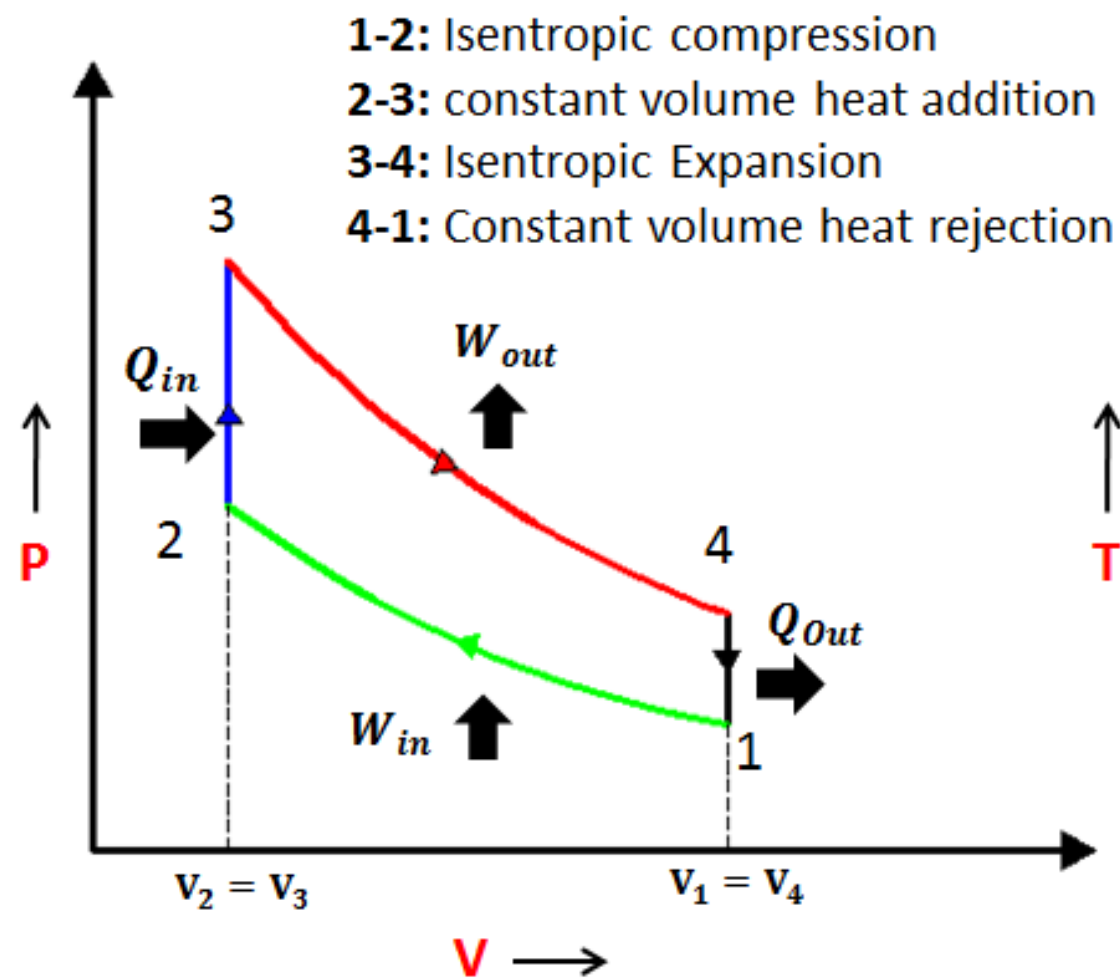
(Aim or what we need actually)

- iv) **Exhaust stroke** - With the exhaust valve open, the piston travels from BDC to TDC in the exhaust stroke.

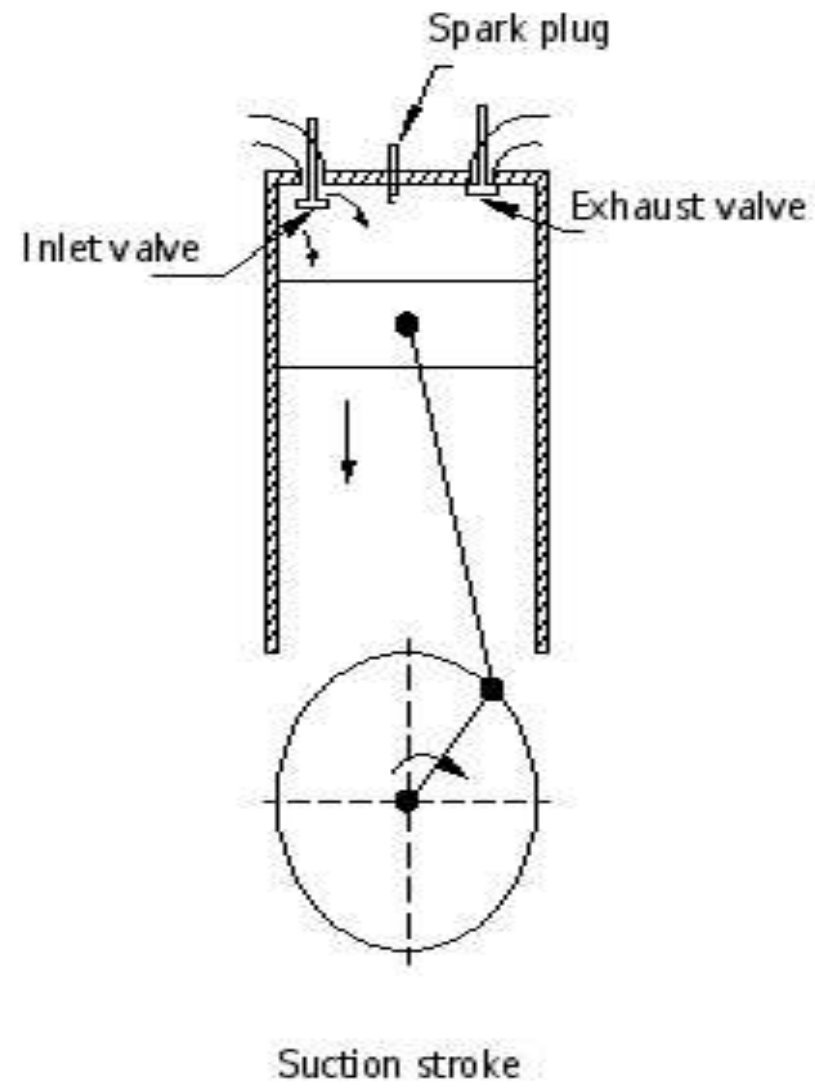
And pushes most of the remaining exhaust gases out.

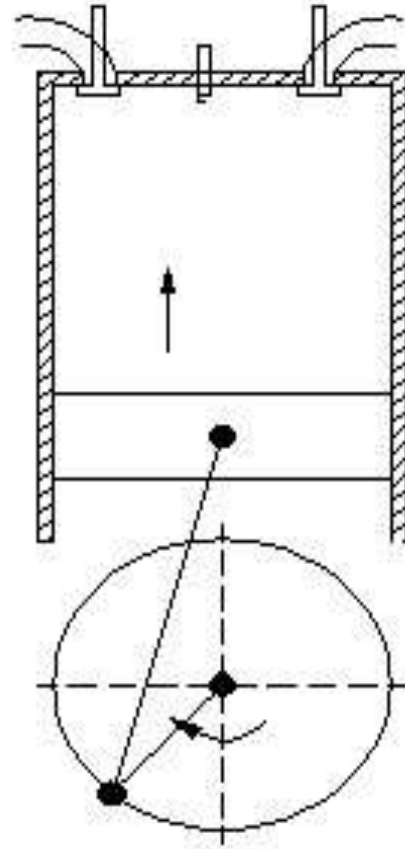


- Petrol engines the compression ratio is generally from 8:1 to 12:1

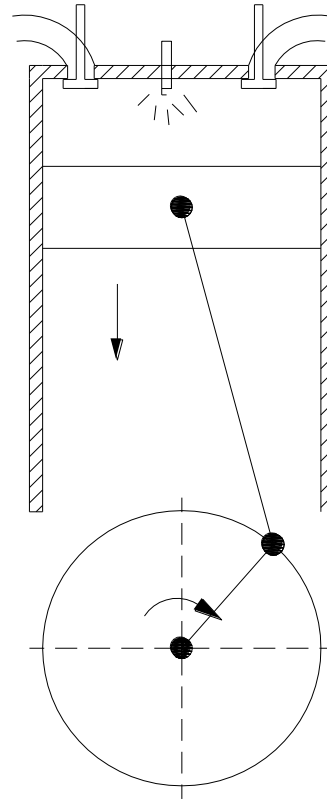


P-V and T-S Diagram of Otto Cycle

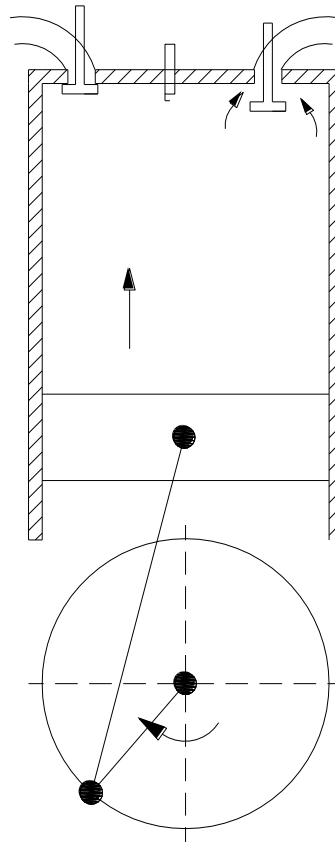




Compression
stroke



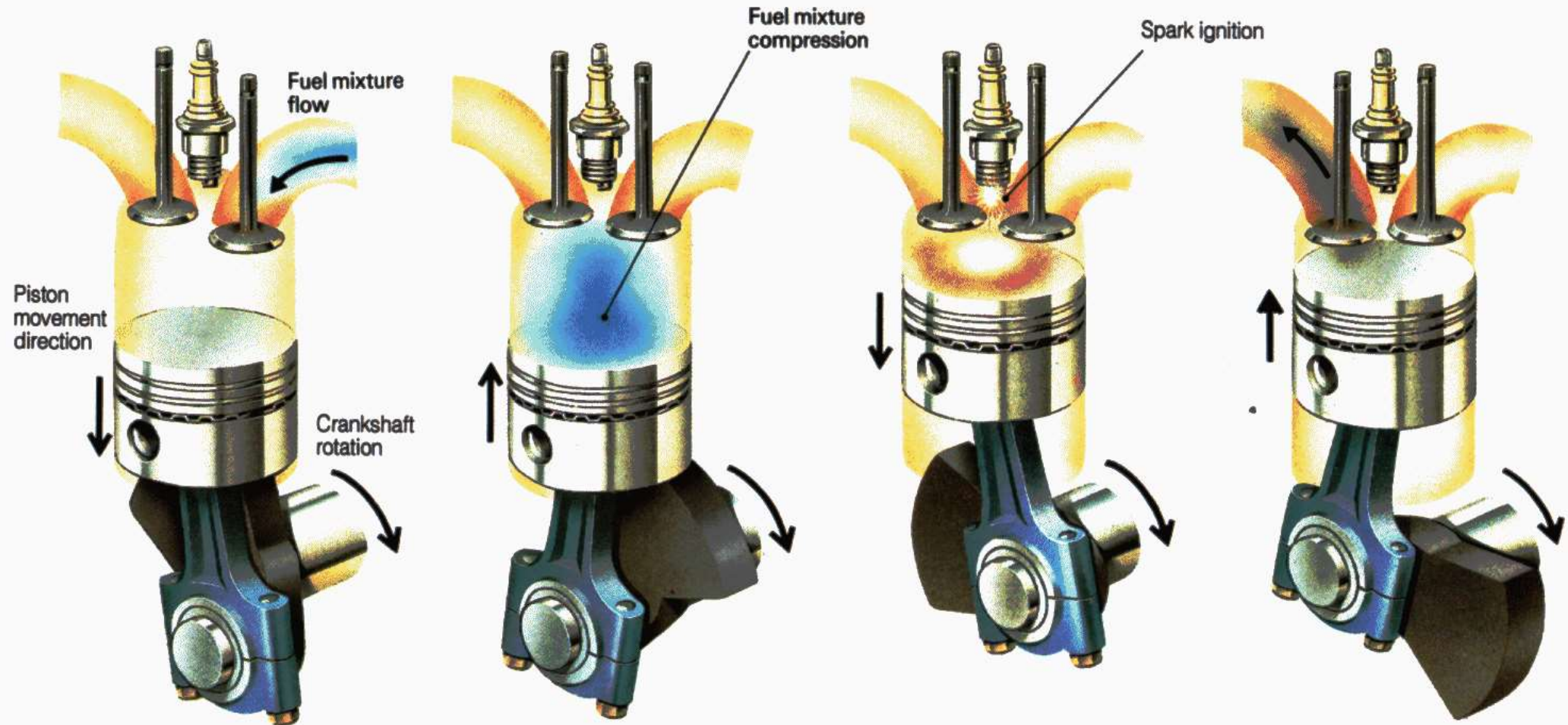
Power stroke



Exhaust stroke

Four stroke SI engine

The four-stroke cycle



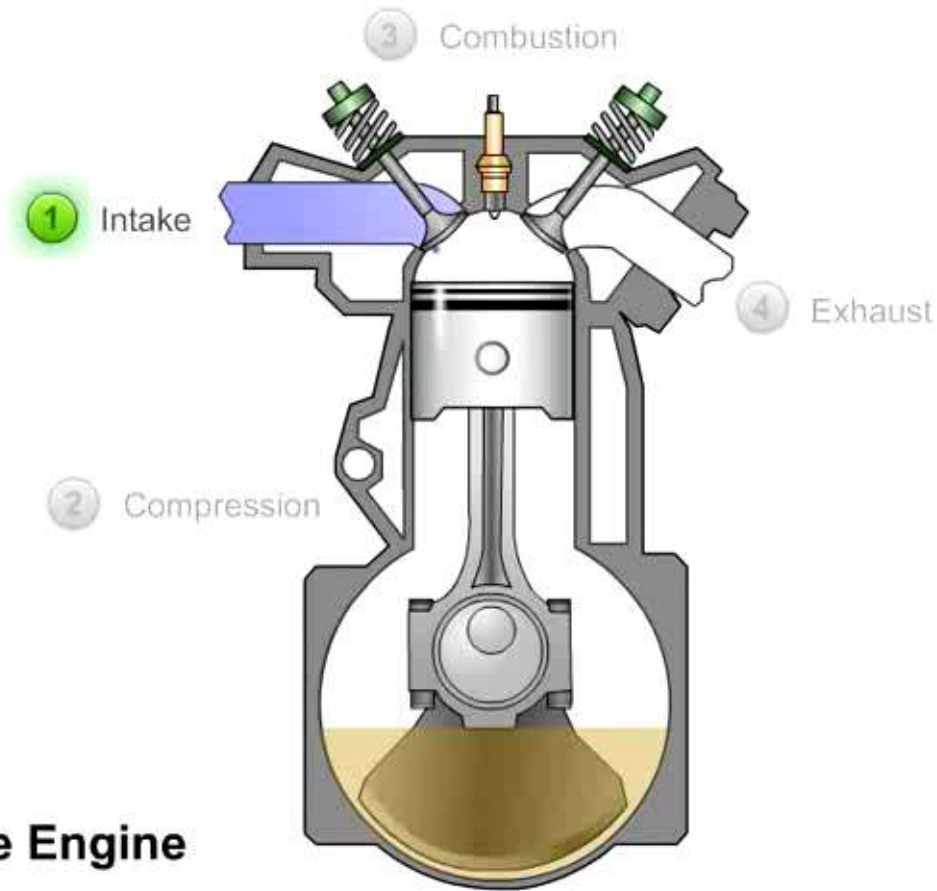
On the induction stroke the piston is descending, the inlet valve is fully open and the exhaust valve closed.

As the piston rises on its compression stroke the exhaust valve is still closed and the inlet valve is closing.

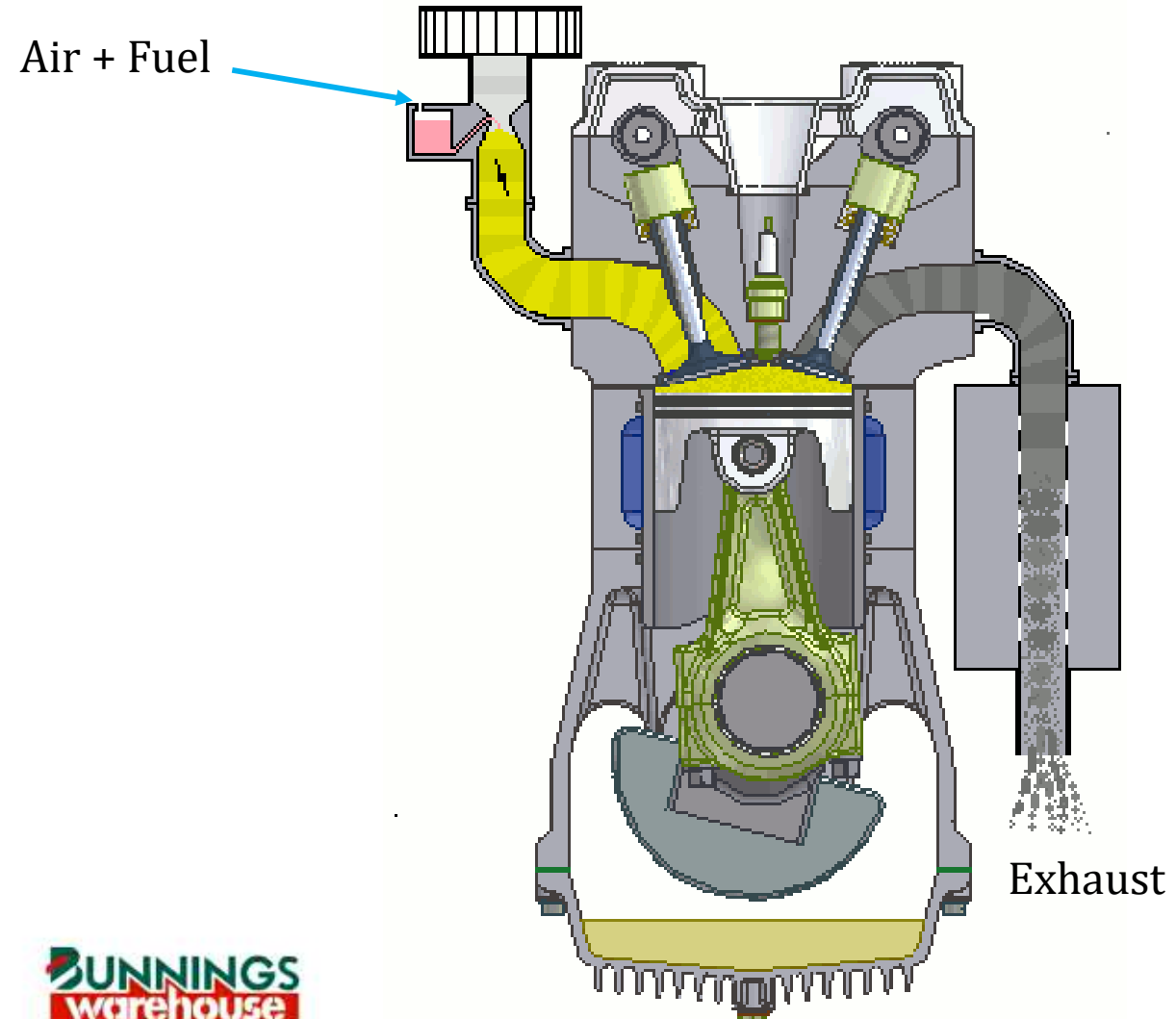
The power stroke drives the piston downwards as the ignited gases expand. Both the inlet and exhaust valves are closed.

The hot gases in the cylinder escape through the open exhaust valve as the piston rises again for the exhaust stroke.

Four stroke – Animation



The 4 Stroke Engine



4 stroke Diesel engine

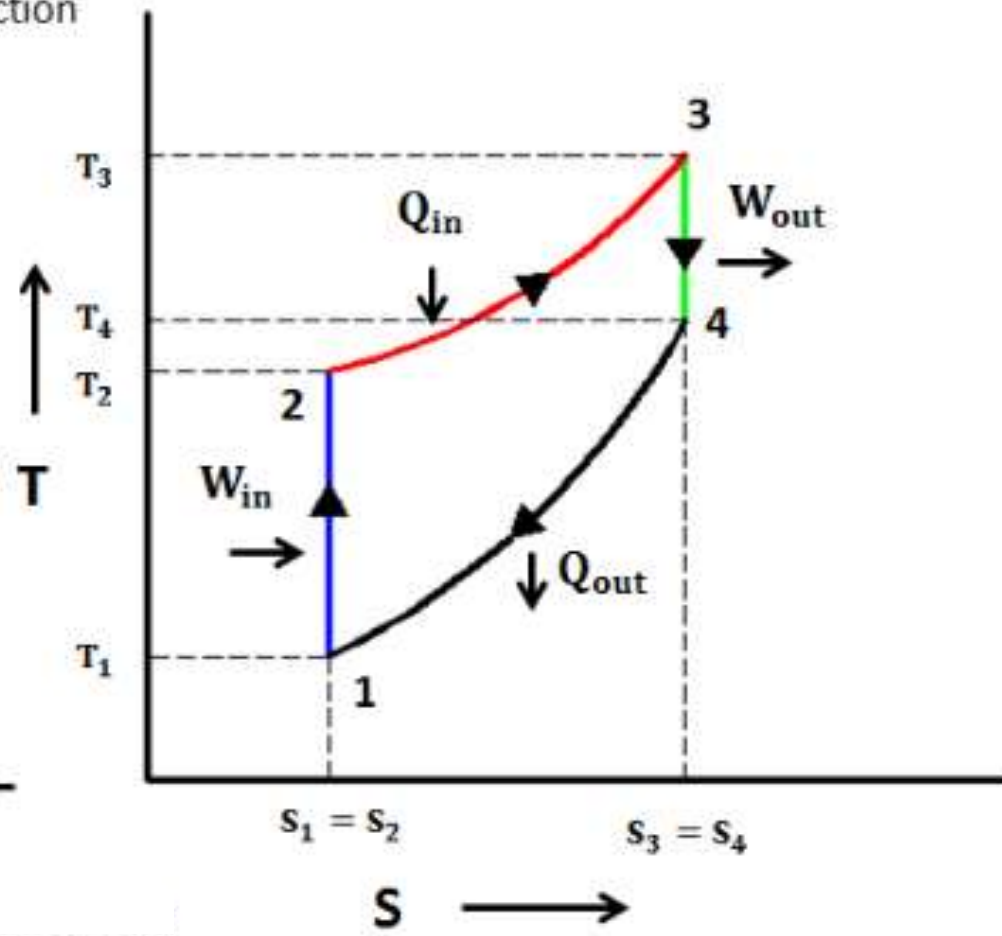
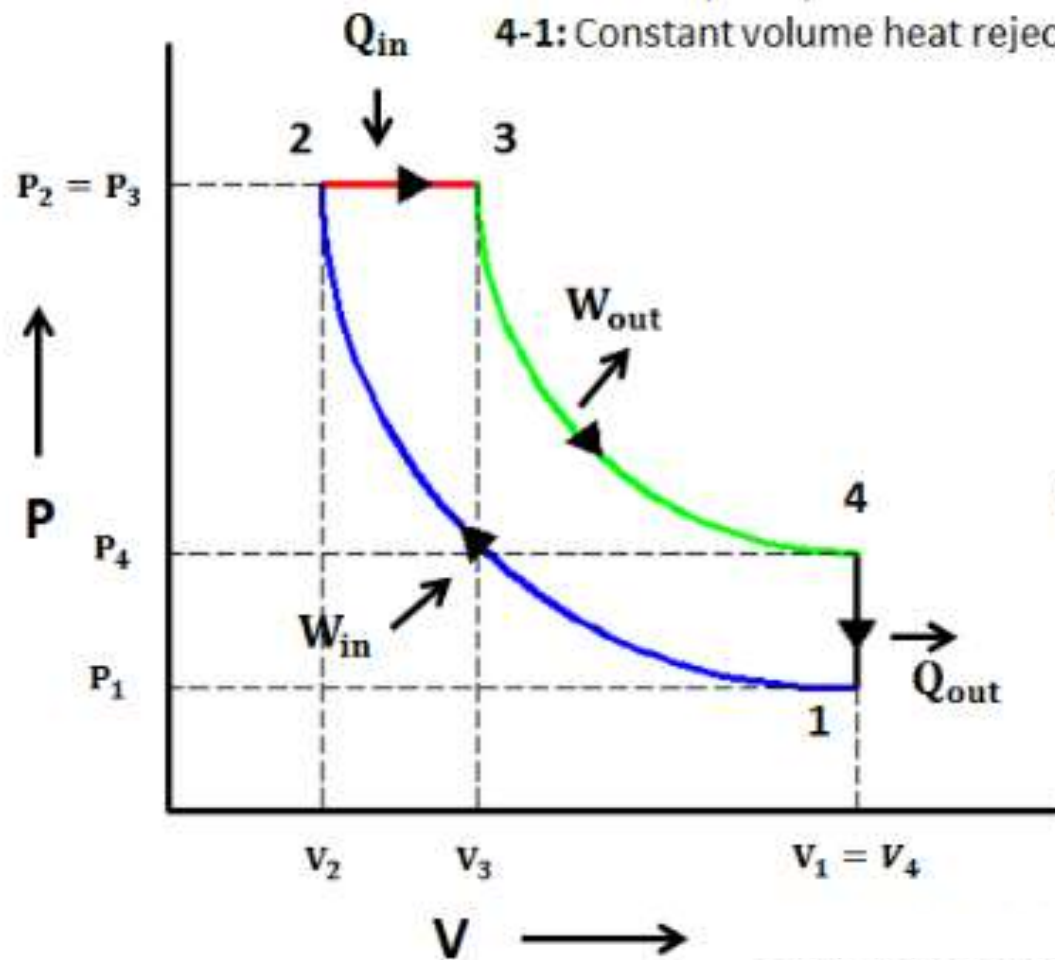
Principle of working.

- Construction schematic is same as SI engine, with the fuel injector (replacing the spark plug) on cylinder head.
- In suction process only air is sucked.
- In compression, the air alone is compressed.
- At the end of compression stroke the temperature of the air inside the cylinder is more than the ignition temperature of diesel.

Contd.

- Now the diesel is injected. After ignition delay period, the combustion starts wherever the A/F ratio is proper for combustion.
- Then flame propagates and combustion completes.
- During this time the piston already starts descending.
- So this combustion may be assumed to be constant-pressure combustion.
- Remaining processes are same as SI engine.
- In CI engines, the compression ratio is generally from 12:1 to 20:1

- 1-2: Isentropic compression
- 2-3: Constant pressure heat addition
- 3-4: Isentropic expansion
- 4-1: Constant volume heat rejection



P-V and T-S Diagram of Diesel Cycle

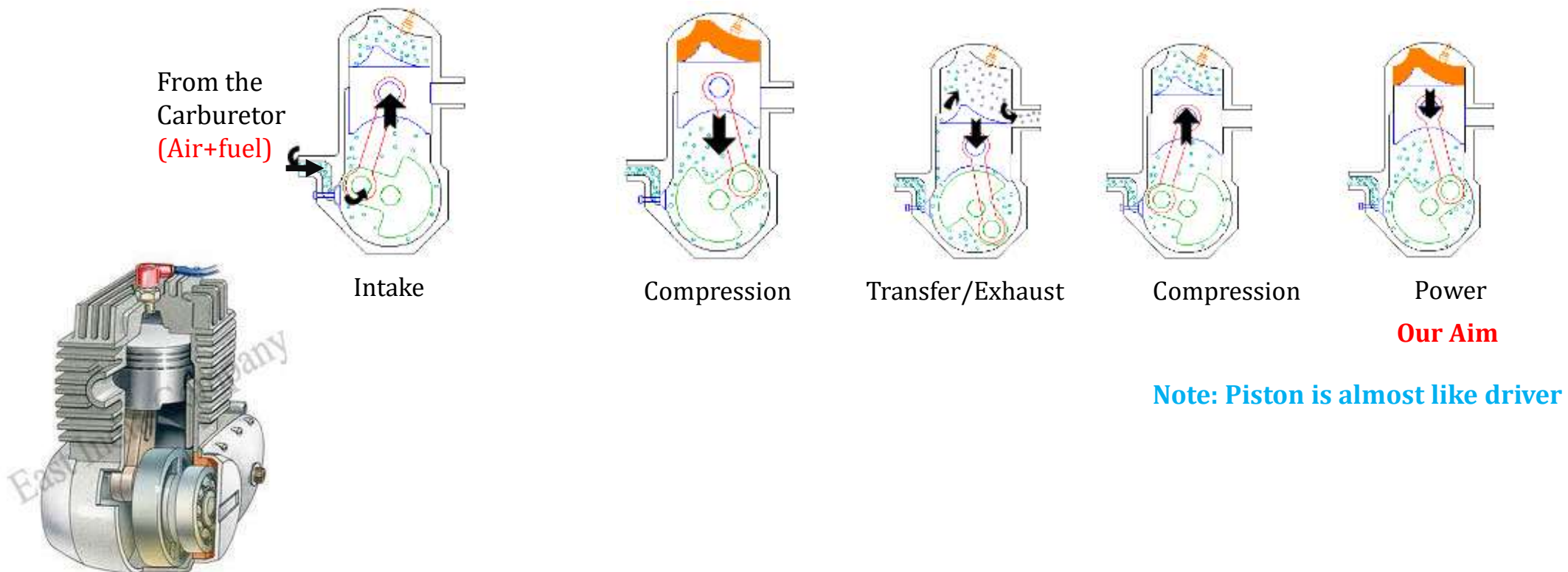
Comparisons of SI and CI engine

S.no	Parameter	SI Engine	CI Engine
1.	Definition	It is an engine in which the spark is used to burn the fuel.	It is an engine in which heat of compressed air is used to burn the fuel.
2.	Fuel used	Petrol and CNG used as fuel.	Diesel is used as fuel.
3.	Operating cycle	It operates on Otto cycle.	It operates on Diesel cycle.
4.	Compression ratio	Low compression ratio (6 to 10)	High compression ratio (15 to 20).
5.	Thermal efficiency	Less thermal efficiency.	High thermal efficiency.
6.	Method of ignition	Spark plug is used to produce spark for the ignition.	Heat of compressed air is used for the ignition.
7.	Engine Speed	High speed engines.	Low speed engines.
8.	Pressure generated	Low pressure is generated after compression.	High pressure is generated after compression.
9.	Intake	Air + fuel.	Only air.
	Weight of engine	SI engine has less weight.	CI engine are heavier.
10.	Noise production	It produces less noise.	It produces more noise.
		.	
11.	Starting	Starting of SI engine is easy.	Starting of CI engine is difficult.
12.	Maintenance cost	Low	High
13.	Vibration problem	low	High
14.	Cost of engine	Less cost	High cost
15.	Fuel supply	Carburetor Or MPFI petrol injection at inlet manifold	Fuel injection Pump and Fuel Injector
16.	application	It is used in light commercial vehicles like motorcycle, cars etc.	It is used in heavy duty vehicles like bus, trucks, ships etc.

Two strokes engine

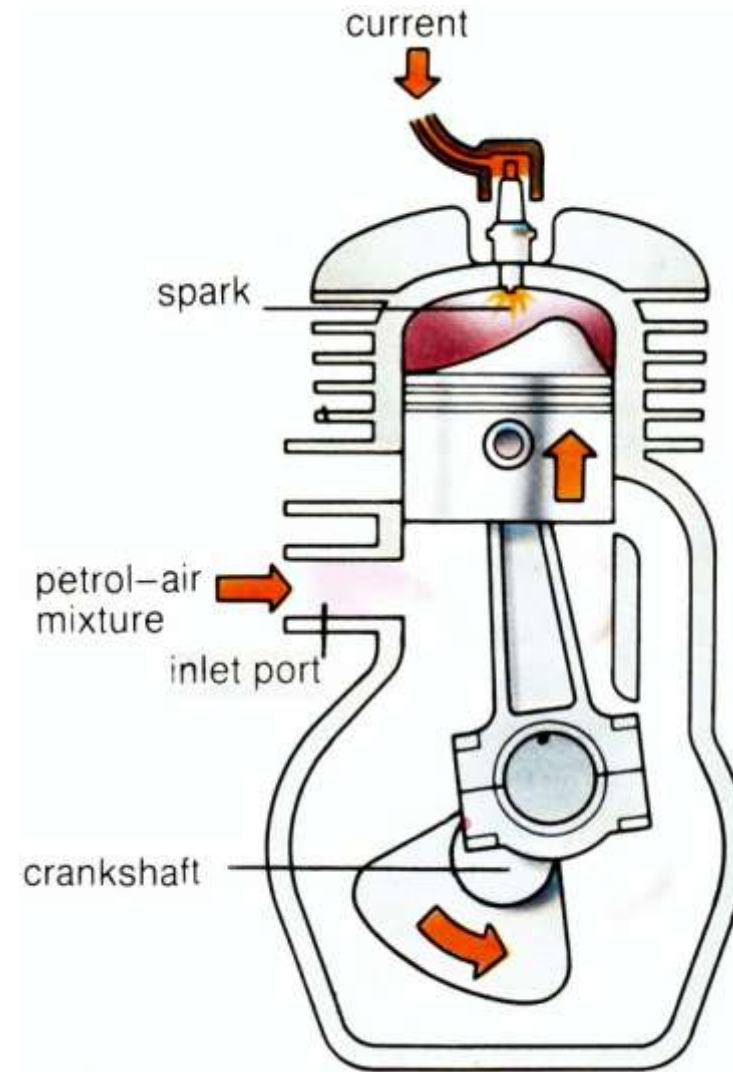
Two stroke SI engine

- SI-Two stroke engines complete the power cycle using an external spark with two strokes of the piston within the cylinder, or one turn of the crankshaft. In these engines, the flow of the intake and the exhaust happen simultaneously.



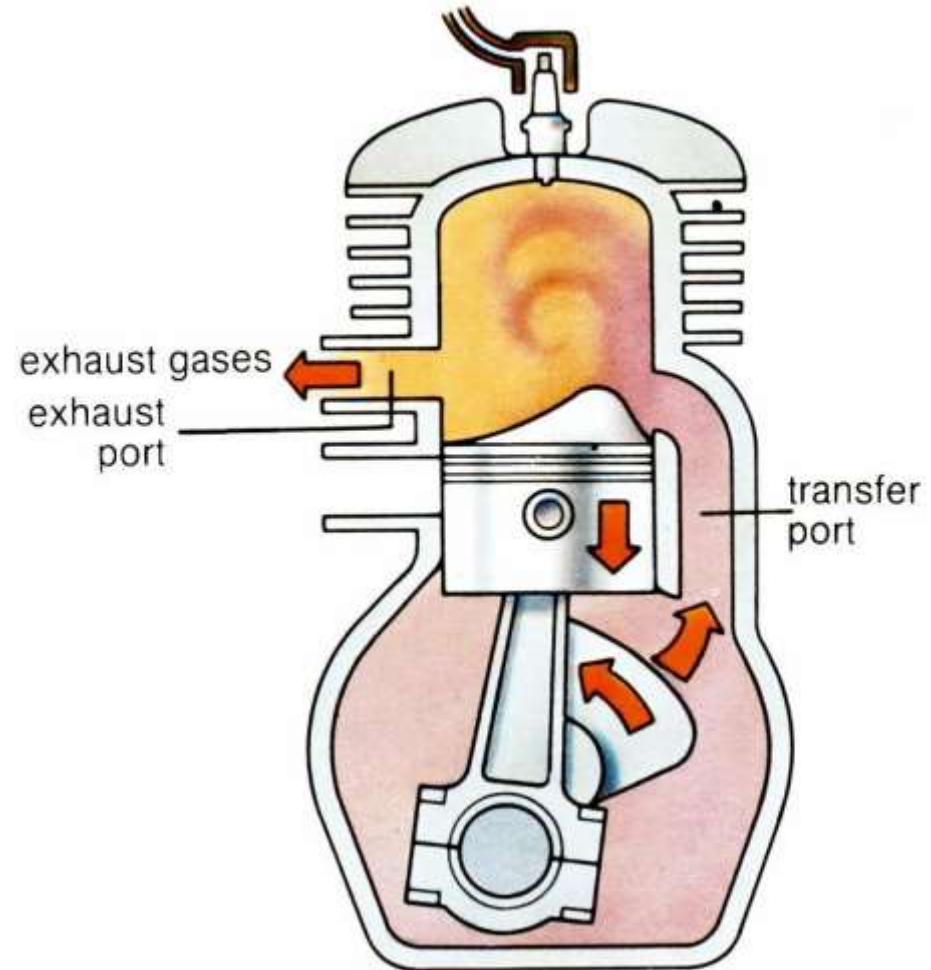
Stroke 1

- The air fuel mixture in the cylinder compressed
- Air fuel mixture enters the crank case through inlet port
- Towards the end of the stroke, the fuel air mixture is ignited using the spark from the spark plug



Stroke 2

- Piston moves downward due to the expansion of the gases
- Near the end of stroke, piston uncovers exhaust port and burnt gases escape through the port.
- Transfer port is uncovered and compressed air fuel mixture from the crankcase flows in to the cylinder



Contd.

- The top profile of the piston helps in deflecting the fresh charge in an upward direction, pushing the remaining gases out.
- Pushing the exhaust gases by the incoming fresh charge is known as scavenging.
- Despite the above process, there is a slight loss of fresh fuel, because of the escape of charge via the exhaust port along with exhaust gases.
- This reduces thermal efficiency, and leads to unburned HC emission.

Advantages

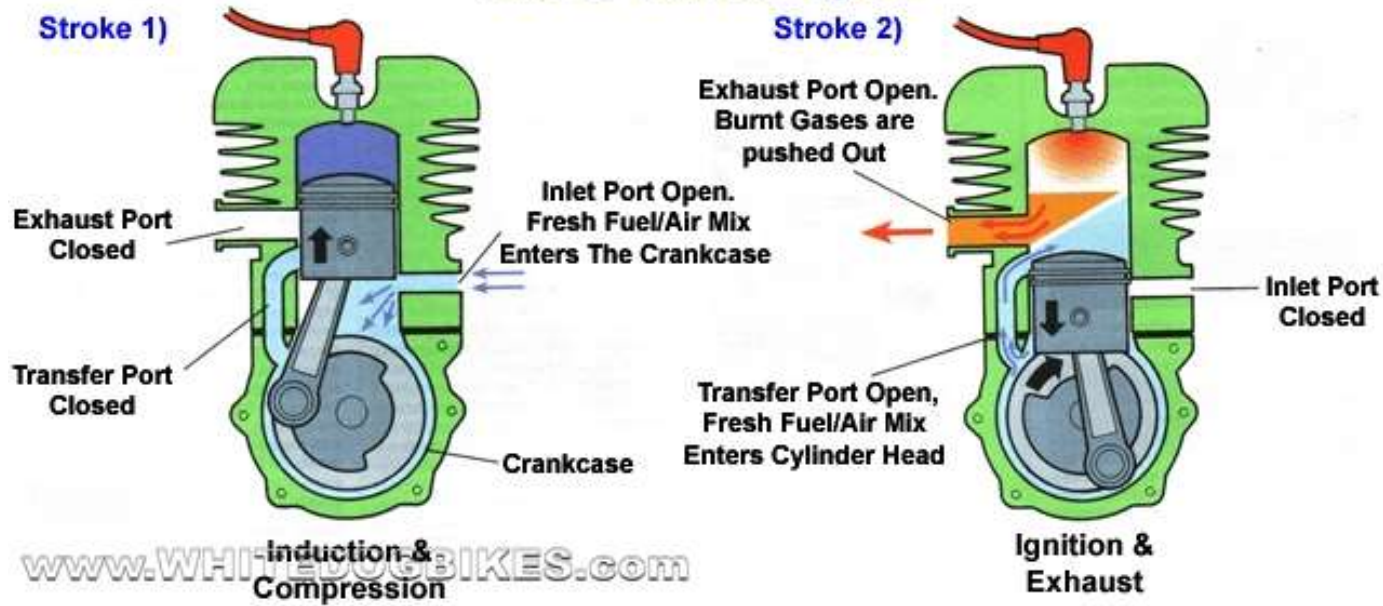
- Two-stroke engines do not have valves which simplifies their construction and lowers their weight.
- Two-stroke engines fire once every revolution, while four-stroke engines fire once every two revolutions. This gives two-stroke engines a significant power boost.
- Theoretically Two-stroke engines develops twice the power into the same space because there are twice as many power strokes per revolution.
- More uniform torque on crank shaft hence it requires a lighter flywheel than that for a four-stroke engine

Disadvantages

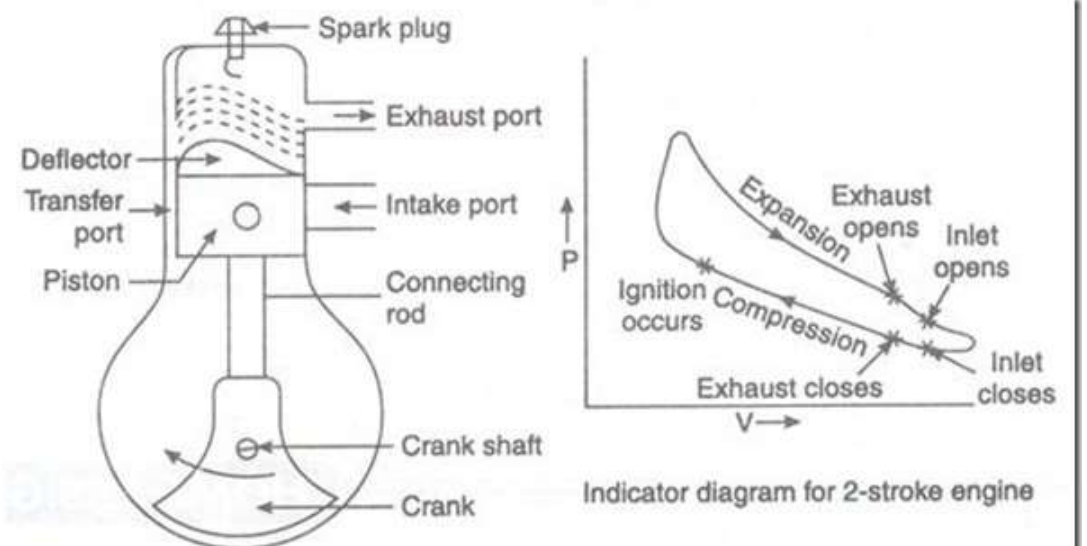
- The engines do not last as long due to poor lubrication.
- You have to mix engine oil with gasoline.
- The engines do not use fuel efficiently.
- These engines produce a lot of pollution.

Two stroke

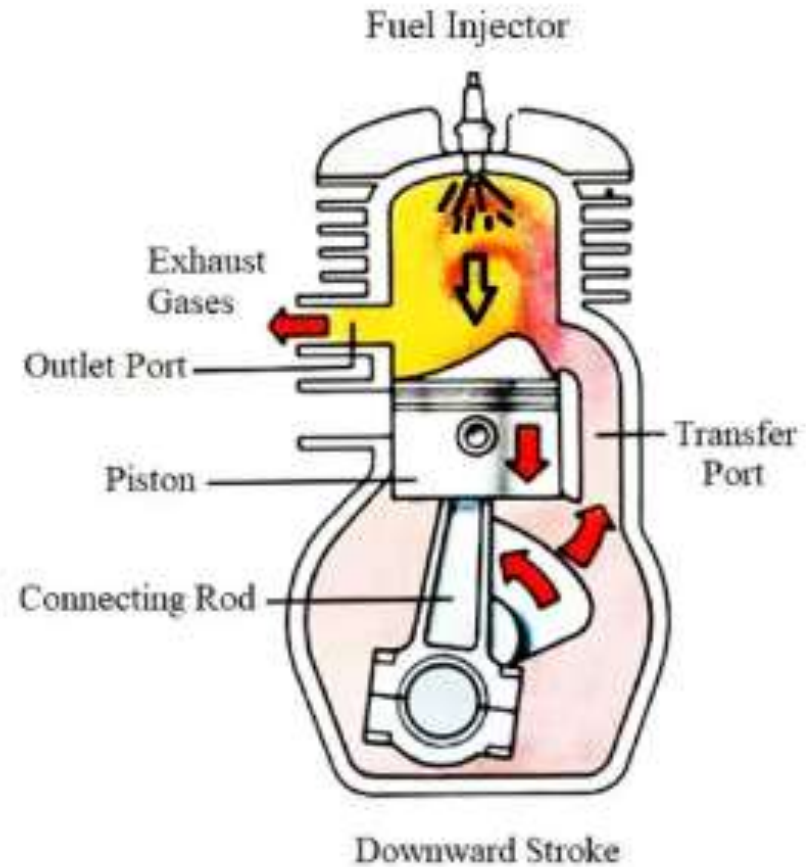
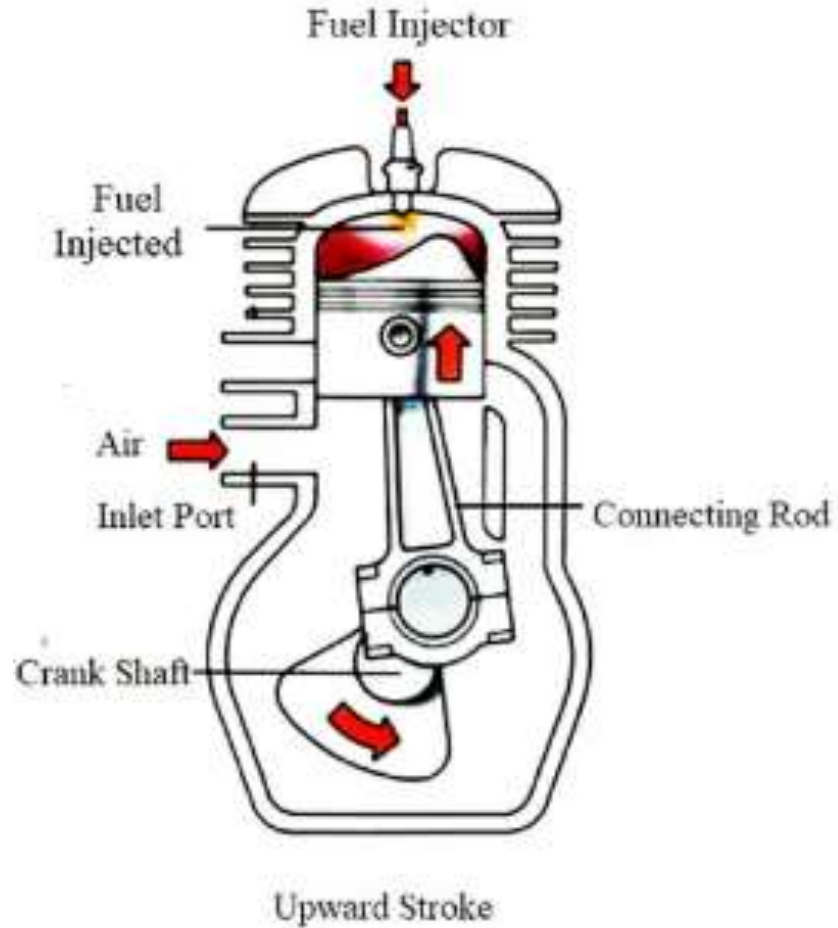
The 2 Stroke Cycle



Crank makes one revaluation, that is 360 degree

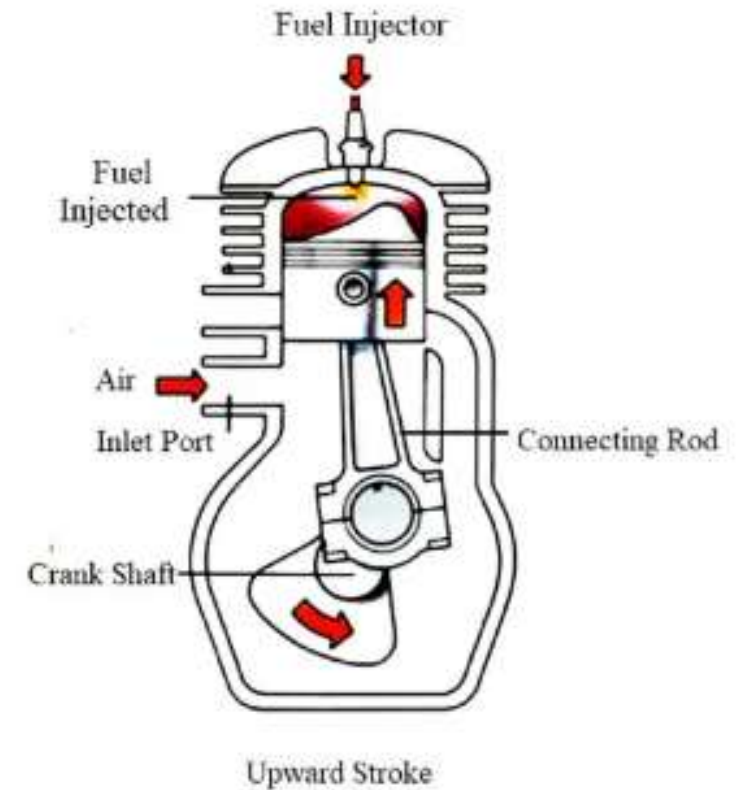


Two stroke Diesel Engine



Two stroke Diesel Engine working.

- Construction schematic is the same as SI engine, with the fuel injector (replacing the spark plug) on the cylinder head.
- Thro' inlet port air only enters into the crank case. Similarly, air enters the cylinder through transfer port.
- When the piston reaches TDC, the temperature of the air inside the cylinder is more than the ignition temperature of diesel.



Contd.

- Now the diesel is injected. After the ignition delay period, the combustion starts wherever the A/F ratio is proper for combustion.
- Then flame propagates and combustion completes.
- Here, there is no fuel loss like in SI engines, because only air comes to the cylinder via the transfer port.

